This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

Cross-Layer Congestion Control Algorithm and Architecture for Communication Performance Optimization in Millimeter-Wave Networks

MIKITO TAKAHASHI[1], AKIHIRO NAKAO[2], (Member, IEEE)
[1]The University of Tokyo (e-mail: mercury00618@gmail.com)
[2]The University of Tokyo (e-mail: nakao@nakao-lab.org)

Corresponding author: Akihiro Nakao (e-mail: nakao@nakao-lab.org).

This work has been partly supported by JST ASPIRE Grant Number JPMJAP2323, Japan

ABSTRACT Increasing mobile data demand and spectrum scarcity have driven 5G/6G networks toward millimeter-wave (mmWave) communication, which offers abundant bandwidth. MmWave links are crucial for emerging applications such as digital twins and artificial intelligence of things (AIoT), requiring ultra-high-speed, low-latency connectivity. However, mmWave signals are highly susceptible to blockage and rapid channel variability, causing frequent non-line-of-sight (NLoS) outages. More importantly, existing communication protocols are primarily designed for stable wired environments and struggle to maintain optimal communication quality in the highly dynamic mmWave environment. In particular, TCP congestion control, which plays a crucial role in mitigating network congestion, fails to adapt to the significant channel variations of mmWave communications, leading to severe performance degradation. To address this issue, we propose an end-to-end cross-layer congestion control architecture that integrates real-time physical-layer channel feedback into transport-layer decisions. The approach classifies NLoS events into weak NLoS (degraded channel with intermittent losses) and strong NLoS (complete link disruption), with tailored algorithms for each case. In weak NLoS, a rapid window adjustment prevents bufferbloat and reduces round-trip time (RTT), while in strong NLoS, transmissions are temporarily halted to avoid futile retransmissions. Simulations show that the proposed method improves throughput by up to 85.7% under strong NLoS blockage and reduces peak RTT by 53.6% in frequent outage (weak NLoS) scenarios, compared to standard TCP. It also maintains fairness across concurrent flows. Importantly, the solution is implemented end-to-end, without the need for support from intermediate network nodes, requiring no changes to existing network infrastructure, making it a cost-effective and easily deployable solution.

INDEX TERMS mmWave, congestion control, TCP, machine learning, flow control, fading model

I. INTRODUCTION

IN recent years, the rapid increase in mobile communication traffic has led to a serious shortage of frequency resources. Moreover, the realization of next-generation social infrastructure technologies such as autonomous driving and smart cities has drawn attention to emerging technologies like digital twins and the artificial intelligence of things (AIoT). These technologies require the development of a high-speed, high-capacity, and low-latency communication infrastructure capable of processing massive amounts of data in real time. Against this backdrop, the utilization of new wireless frequency bands (New Radio: NR) has been actively considered in 5G and future wireless communication standards such as 6G and beyond. In particular, millimeter-wave (mmWave) communication, which offers a large available bandwidth and significantly higher communication capacity than conventional wireless technologies, has garnered considerable attention.

However, it is a well-known fact that mmWave communication has certain high-frequency characteristics, such as high susceptibility to scattering and low diffraction capability, which lead to frequent disruption of communication links due to obstacles and increased signal interference from multipath propagation. These phenomena result in two major problems: (1) an inability to transmit data effectively even under favorable communication conditions, and (2) a degradation of

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

communication quality caused by forced data transmission under poor channel conditions.

On the other hand, we observe that standard TCP congestion control performs poorly over mmWave links, especially in rapidly changing non-line-of-sight (NLoS) environments. As a mmWave channel shifts from line-of-sight (LoS) to NLoS (or vice versa), the data rate can plunge or vanish within milliseconds. Conventional TCP mistakes these wireless impairments for congestion: it cuts the congestion window (cwnd), incurs retransmission timeouts, and then slowly ramps up again. The result is underutilized bandwidth when the channel recovers, and large queuing delays (bufferbloat) during fades. Crucially, not all NLoS fades are equal: some cause total blockage, which we define as "strong NLoS", while others only partially attenuate the signal, which we call "weak NLoS", leading to frequent retransmissions. Off-the-shelf TCP treats both cases alike, which either over-backs off or wastes effort. In summary, mmWave flows experience significant throughput loss and latency inflation under today's TCP schemes, highlighting the need for specialized solutions.

To address these challenges, this paper proposes a cross-layer framework focused on redesigning congestion control algorithms. By designing congestion control mechanisms optimized for mmWave environments, this study aims to maximize the potential of mmWave communication and establish a high-speed, high-capacity, and low-latency communication infrastructure. Such an infrastructure is expected to serve as a fundamental foundation for the realization of autonomous driving, remote healthcare, and smart cities.

Our key ideas are fourfold as follows:

- *NLoS classification:* We rigorously define two distinct NLoS regimes – weak NLoS (moderate fading with recoverable signal-to-interference-plus-noise ratio (SINR)) and strong NLoS (deep fades or blockages causing near-zero capacity) based on physical channel behavior. This enables tailored control strategies for each scenario.
- *Weak NLoS congestion control:* We design a lightweight receiver-side module that quickly detects moderate SINR drops and smoothly adjusts the cwnd. This avoids TCP's harsh back-offs in mild fades. In simulations, our weak NLoS algorithm reduces the maximum round-trip time (RTT) by 53.6% and improves throughput by 9.8% compared to unmodified TCP CUBIC.
- *Strong NLoS congestion control:* For deep fades, we develop a robust fallback mechanism: upon detecting total loss of LoS, the sender enters a conservative hold state rather than blindly retransmitting. Once the channel recovers, the window resumes promptly. This approach yields dramatic gains under blockage, with up to 85.7% higher throughput than CUBIC in single-flow tests.
- *Low-cost cross-layer architecture:* We implement these algorithms in an end-to-end, simple cross-layer fashion, inspired by adaptive receive window adjustment frameworks. A few added modules feed timely channel state information into the TCP stack. This design requires minimal protocol changes and leverages existing hardware, making it practical for deployment. Specifically, in the weak NLoS approach, only a lightweight cross-layer mechanism is needed within the mobile device: lower-layer modules provide SINR information, and the transport layer incorporates a simple learning mechanism to interpret wireless channel states. No modification to the server or the network infrastructure is required. In the strong NLoS approach, additional implementation on the server side is required to support explicit feedback, but it can be integrated with minimal changes by augmenting the existing CUBIC congestion control algorithm with an RTO suppression mechanism. Importantly, both approaches do not rely on complex and computationally intensive algorithms such as deep learning or reinforcement learning. Moreover, they do not require any support or modification from the network side, making them lightweight and cost-effective architectures. As discussed in Section VII, all processing is performed locally within the mobile device (except for the optional server-side module in strong NLoS), maintaining full backward compatibility with existing TCP implementations and network infrastructure. This backward compatibility and minimal implementation effort enhance deployability, resulting in a practical and low-cost cross-layer architecture.

Altogether, these innovations create a protocol stack that is NLoS-aware and adapts in real time to mmWave fading. Our extensive ns-3 simulations show that the proposed scheme maintains high link utilization and low latency under both weak and strong blockages, far outperforming standard TCP. In particular, the algorithm for strong NLoS achieves 85.7% throughput gain in blocked scenarios, while the weak NLoS tweak prevents bufferbloat under mild fades.

In summary, we deliver a novel cross-layer congestion control architecture that (i) explicitly identifies NLoS severity, (ii) applies tailored TCP adjustments for each case, and (iii) is readily implementable in current networks. This approach substantially boosts end-to-end performance over mmWave, addressing the critical gap between transport protocols and the reality of high-frequency wireless channels.

Prior approaches tackle mmWave TCP issues in different ways. Machine learning methods, such as fuzzy logic or deep neural controllers, can learn complex channel patterns but incur training overhead and unpredictability. Proxy architectures (e.g., milliProxy) improve performance without modifying end hosts, but require new network elements. Other cross-layer schemes like X-TCP use PHY rate estimates to set cwnd, often focusing on uplinks or synchronizing with base-station scheduling. Receiver-side window-adjustment (DRWA/ABRWDA) methods simply shrink the advertised window based on RTT/SINR, but treat all fades alike (see Section III). In contrast, our design explicitly distinguishes weak vs. strong fades and applies appropriate TCP control in each case. Unlike proxies or machine learning models, our solution works at the mobile terminal without extra infrastructure and

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

with minimal computation. It builds on the idea of cross-layer adaptation, but goes further by segmenting NLoS behavior and using targeted algorithms, achieving higher robustness and simplicity than existing schemes.

The remainder of this paper is structured as follows. Section II defines the problem statement this paper addresses. Section III introduces related work on congestion control in mmWave environments and highlights the differences from the proposed approach. Section IV presents the proposed cross-layer congestion control algorithms tailored for weak and strong NLoS environments. Section V evaluates the proposed algorithms using ns-3 simulations and verifies their effectiveness. Finally, Section VIII concludes this paper.

II. PROBLEM STATEMENT

Congestion control is a core technology that determines the communication stability in environments where numerous terminals are simultaneously communicating. However, conventional congestion control algorithms are designed under the assumption that network conditions change gradually, which holds true for wired environments. In contrast, in mmWave environments where network conditions vary rapidly, conventional algorithms fail to operate effectively. In particular, compared to LoS environments, where direct wave propagation is possible without obstacles, NLoS environments, where direct wave propagation is blocked by obstacles, exhibit abrupt wireless channel variations. These rapid variations make it difficult to appropriately control the cwnd, leading to a decline in throughput and increased bufferbloat. Consequently, communication quality in mmWave networks can significantly deteriorate compared to existing wireless communication systems.

A primary cause of these problems is that the existing communication protocol stack is designed based on the characteristics of wired environments and low-frequency bands, without sufficient consideration for the unique properties of mmWave communication. Therefore, to fully exploit the advantages of mmWave communication, it is essential to redesign the entire communication protocol stack to accommodate the mmWave environment. In particular, the redesign of congestion control, which plays a crucial role in mitigating communication congestion, is of paramount importance.

Several congestion control algorithms have been proposed to address the degradation of communication quality in NLoS environments. However, the underlying mechanisms causing these issues and their impact on the entire protocol stack have not been adequately addressed. As a result, existing proposals have been limited to addressing specific problems rather than fundamentally improving communication quality. Furthermore, for mmWave communication to be widely deployed in real-world scenarios, the proposed algorithms must be implemented in a simple and cost-effective manner. However, discussions on the architectural design of such low-cost implementations remain insufficient.

In the light of these observations, we posit that to promote the widespread adoption of mmWave communication, it is essential to (1) systematically identify the factors that cause communication quality degradation in mmWave environments and design congestion control algorithms that address the fundamental nature of these problems, and (2) develop an architecture that allows the proposed algorithms to be deployed in a cost-effective manner.

A. CHALLENGES RELATED TO COMMUNICATION QUALITY

The primary challenges related to communication quality include inappropriate Retransmission Timeout (RTO) occurrences in congestion control and excessive latency caused by bufferbloat.

1) **Inappropriate RTO Occurrences**: In mmWave communication, data rate degradation frequently occurs in NLoS environments, increasing the likelihood of packet loss. However, traditional TCP congestion control mistakenly interprets these packet losses as network congestion, triggering unnecessary RTO events and reducing the cwnd size. As a result, even after recovery to an LoS environment, it takes considerable time to restore the appropriate window size, causing significant throughput degradation.
2) **Bufferbloat**: Excessive buffering and retransmission mechanisms temporarily conceal the effects of channel variations; however, this leads to bufferbloat, increasing communication latency. Consequently, the network state cannot be accurately reflected in real-time, resulting in improper operation of congestion control.

B. BARRIERS TO SOCIAL IMPLEMENTATION

The key barriers to social implementation are the high cost of deploying advanced communication enhancement technologies and compatibility issues with existing communication infrastructure and systems.

1) **High Deployment Cost**: Adaptive control techniques based on deep learning and network-assisted control methods are expected to improve communication reliability and throughput significantly. However, implementing these methods involves substantial time and financial costs, which act as barriers to adoption.
2) **Compatibility with Existing Infrastructure**: Algorithms and architectures specifically designed for mmWave communication often have low compatibility with existing network infrastructure. This makes implementation challenging and hinders seamless migration.

III. RELATED WORK

In mmWave communication environments, it has been shown that traditional TCP congestion control mistakenly interprets large link variations associated with LoS and NLoS transitions as congestion, leading to significant degradation in communication quality [1]. Therefore, designing new congestion control mechanisms adapted to mmWave communication environments has become a critical challenge.

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

A. MACHINE LEARNING-BASED APPROACHES

To improve traditional congestion control, approaches based on fuzzy control and deep learning have been proposed [2], [3]. These methods employ more sophisticated control mechanisms, thereby improving communication quality in mmWave wireless channel environments. In particular, the application of highly expressive algorithms such as machine learning and deep learning has demonstrated the potential to adapt to environmental variations that are difficult to handle with traditional methods.

B. CROSS-LAYER APPROACHES

Cross-layer approaches that utilize information from the physical layer, which is below the transport layer, have also been proposed [4], [5]. Cross-layer approaches involve redesigning the interfaces between different modules in the Open Systems Interconnection reference model, enabling information sharing across module boundaries. By adopting such an architecture, it becomes possible to accurately capture wireless channel conditions and apply this information for more effective control.

C. NETWORK-ASSISTED CONGESTION CONTROL

Another approach involves leveraging base stations or proxies to assist in congestion control across the entire network topology [6], [7]. This method has shown to be highly effective in improving communication quality by supporting congestion control at the network level. However, the high deployment cost associated with modifying base stations or installing new proxies presents a significant barrier to practical implementation.

In addition, there exist methods that aim to enhance communication performance by utilizing multiple communication paths simultaneously, such as Multipath TCP (MPTCP) [8]. While these approaches are attracting attention as promising solutions for achieving more reliable communication in mmWave mobile networks, existing variants have not yet demonstrated sufficient robustness in improving performance under highly variable mmWave conditions. Therefore, the development of new protocol designs remains an open challenge.

D. ACTIVE QUEUE MANAGEMENT

An alternative approach involves coordinating congestion control with Active Queue Management (AQM) [9], [10]. In particular, Multi-Agent Congestion Control (MACC) [11] employs deep reinforcement learning-based agents at the transport and data link layers, enabling protocol-level coordination. This approach has demonstrated improved communication quality compared to conventional combinations of AQM and congestion control. The novelty of this method lies in its ability to enable cross-layer coordination, which is promising given the increasing complexity and microservice-based nature of modern communication systems.

E. IMPLICIT CROSS-LAYER AND NETWORK-AWARE APPROACH

The method proposed in [12] is an RTT-based congestion control scheme designed with a focus on efficient bandwidth utilization in mmWave LoS environments. Although the algorithm does not explicitly incorporate cross-layer mechanisms, it implicitly achieves a cross-layer design by leveraging the bandwidth-delay product (BDP) and considering the buffer occupancy status within the network. These factors are collectively reflected through variations in RTT, making the approach both implicitly cross-layer and network-aware.

F. DYNAMIC RECEIVE WINDOW ADJUSTMENT

To reduce bufferbloat in mmWave environments while minimizing deployment costs, Dynamic Receive Window Adjustment (DRWA) has been proposed [13]. DRWA adjusts the receive window (rwnd) dynamically based on RTT using flow control mechanisms that regulate data transmission speed between the sender and receiver. This approach has been shown to reduce RTT and improve throughput in high BDP environments. A key advantage of DRWA is that it can be implemented by modifying only the receiver-side algorithm, preserving backward compatibility with existing network infrastructure. Consequently, communication quality improvements can be achieved with minimal impact on existing network devices.

However, DRWA faces a challenge in that estimation errors in RTT and cwnd size can degrade communication quality compared to delay-based congestion control algorithms such as TCP Vegas [14]. To address this issue, Adaptive Bandwidth-based Receive Window Adjustment (ABRWDA) has been proposed, which recalculates the BDP using SINR information and RTT, and adjusts the rwnd accordingly [15]. ABRWDA has demonstrated significant improvements in delay characteristics compared to conventional DRWA.

ABRWDA is a highly suitable architecture for mmWave communication due to its ease of deployment and the stable communication quality improvements enabled by cross-layer integration. This combination makes ABRWDA a promising solution for promoting the widespread adoption of mmWave communication.

G. POSITIONING OF THIS WORK IN THE CONTEXT OF EXISTING RESEARCH

We summarize the features and novelties of our proposed methods compared to existing related works.

The proposed method for weak NLoS is implemented exclusively by modifying the receiver-side rwnd, similar to DRWA and ABRWDA. In particular, it shares a cross-layer design philosophy with ABRWDA. However, while ABRWDA determines the rwnd value using the BDP, which is calculated based on both direct SINR measurements and RTT, the proposed method introduces a novel mechanism. It learns the underlying SINR distribution through variational Bayesian inference and estimates the appropriate rwnd via a regression model that utilizes the predicted SINR and

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

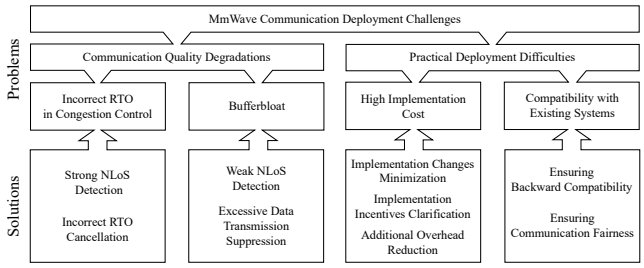

FIGURE 1. Degradation of Communication Quality and Barriers to Social Implementation in Millimeter-Wave Communication and Their Respective Solutions.

previously observed throughput data. This approach enables efficient learning of key characteristics of the mmWave environment from the model distribution and maintains fairness with traditional loss-based congestion control schemes.

The proposed method for strong NLoS adopts an architecture implemented on both the sender and receiver sides. Compared to methods that modify the algorithm on only one side, either sender or receiver, this approach entails a higher deployment cost. However, it is considered to have a lower deployment cost than schemes that rely on congestion control assistance at base stations or proxies. A key novelty of the proposed method is its use of Bayesian learning to model the SINR distribution, which enables suppression of RTOs when severe signal degradation is detected. By preventing cwnd resets in environments experiencing significant signal drops, this approach can achieve marked throughput improvements.

A comprehensive comparison between the related works reviewed in this section and the two proposed methods of this study is summarized in Table 1.

IV. PROPOSED METHOD

To address the issues defined in our problem statement in Section II, Communication Quality Degradations and Practical Deployment Difficulties, we propose the solutions to these challenges as depicted in Fig. 1.

To address the challenges related to communication quality, different methods should be applied based on the severity of the environmental impact. Specifically, the communication environment should be classified into a "**strong NLoS Environment**" with strong NLoS effects and a "**weak NLoS Environment**" with relatively mild NLoS effects. We classify the NLoS environment into strong NLoS and weak NLoS environments and design appropriate congestion control methods for each. Specifically, we model the SINR distribution for each environment and enable dynamic prediction of channel states. This prediction model is then applied to congestion control, and by adopting a cross-layer approach using SINR, we achieve control that adapts to the complex fluctuations of millimeter-wave wireless channels.

1) **Strong NLoS Environment**: In strong NLoS environments, real-time detection and cancellation of unnecessary RTO events should be introduced to prevent excessive reductions in the cwnd size.
2) **Weak NLoS Environment**: In weak NLoS environments, mechanisms to suppress excessive data transmission should be implemented to mitigate bufferbloat and reduce communication latency.

To overcome barriers to social implementation, reducing deployment costs and ensuring compatibility with existing systems are essential. Regarding deployment costs, minimizing the number of implementation points while clarifying the benefits of deployment and reducing additional overhead will enhance cost-effectiveness. Furthermore, to ensure compatibility with existing systems, backward compatibility should be maintained, and fair congestion control across new and existing systems should be achieved to facilitate smooth migration.

By implementing these measures, the adoption of mmWave wireless communication is expected to accelerate, leading to the establishment of a high-quality and highly practical communication environment.

A. PHYSICAL LAYER MODELING

In this section, we model the SINR distribution for strong NLoS and weak NLoS environments, which are required for channel estimation.

In millimeter-wave communications, due to the high frequency and high reflectivity characteristics, signals reach terminals via multipath propagation. The fading model, including both direct and reflected waves, is formulated by the Rician distribution in Eq. 1 [16].

$$f(x;a,\sigma) = \frac{x}{\sigma^2}\exp\left(-\frac{x^2+a^2}{2\sigma^2}\right) I_0\left(\frac{ax}{\sigma^2}\right), \quad x \geq 0 \quad (1)$$

The variables used in Eq. 1 are as follows:

- x: Observed signal strength
- a: Direct wave component
- σ: Noise component

Additionally, $I_0(z)$ is the modified Bessel function of the first kind, expressed by Eq. 2.

$$I_0(z) = \sum_{k=0}^{\infty} \frac{1}{(k!)^2}\left(\frac{z}{2}\right)^{2k} \quad (2)$$

When a is much larger than the noise component σ, this distribution corresponds to the signal strength distribution in the LoS environment. On the other hand, as a becomes smaller, the effect of noise gradually increases, and the distribution represents a weak NLoS environment where the signal strength is reduced. When $a = 0$, this distribution becomes the Rayleigh distribution, as shown in Eq. 3, representing a strong NLoS environment where no direct wave exists.

$$f(x;\sigma) = \begin{cases} \frac{x}{\sigma^2}\exp\left(-\frac{x^2}{2\sigma^2}\right) & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (3)$$

In the proposed method of this paper, since SINR is used as a physical layer parameter, it is necessary to transform the distributions in Eqs. 1 and 3 into SINR distributions.

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

TABLE 1. Summary of Congestion Control Approaches in mmWave Mobile Networks

Approach	Related Works	Key Idea	Advantages	Limitations
Machine Learning-Based	[2], [3]	Apply fuzzy logic and deep learning for adaptive congestion control	High adaptability to channel variations; potential for automation	High computational cost; requires large training data
Cross-Layer Design	[4], [5]	Use lower-layer information (e.g., PHY layer) to improve TCP responsiveness	Accurate channel awareness; improves throughput	Requires redesign of protocol interfaces
Network-Assisted Control	[6], [7]	Utilize base stations or proxies to assist TCP control	Network-wide optimization; robust performance	High deployment and operational cost
Multipath Communication	[8]	Leverage multiple paths for redundancy and improved reliability	Resilient to blockage and link degradation	Existing variants lack robustness in mmWave; needs protocol enhancement
AQM	[9]–[11]	Combine AQM and congestion control; use multi-agent learning	Cross-layer protocol coordination; good quality improvement	Implementation complexity; learning stability
Implicit Cross-Layer and Network-Aware	[12]	Use RTT variation to implicitly reflect buffer and bandwidth state	No explicit layer modification; adaptable to LoS changes	Communication performance depends on buffer size
DRWA	[13]	Adjust rwnd based on RTT to avoid bufferbloat	Low deployment cost; backward compatibility	Sensitive to estimation errors in RTT and cwnd
ABRWDA	[15]	Use SINR + RTT to recalculate BDP for precise rwnd control	Excellent delay performance; cross-layer benefits	Relies on accurate SINR estimation; moderate complexity
Proposed Method for Weak NLoS	This work	Learn SINR distribution via variational Bayesian inference and detect channel changes to set appropriate rwnd	Low deployment cost; mitigates bufferbloat; improves communication quality while maintaining fairness	Requires SINR monitoring and parameter tuning for stable learning
Proposed Method for Strong NLoS	This work	Learn SINR distribution via Bayesian inference to detect severe signal degradation and avoid inappropriate RTOs	End-to-end implementation; significantly improves throughput	Requires SINR monitoring; additional implementation at both sender and receiver

First, we calculate the SINR distribution in the weak NLoS environment. Direct transformation from the Rician distribution results in a distribution that is difficult to handle when applying Bayesian learning algorithms. To solve this problem, we approximate the Rician distribution with a Gaussian distribution, which is an asymptotic distribution. By transforming this Gaussian distribution, we obtain a Gaussian-Gamma distribution as shown in Eq. 4.

$$f(z; a, \gamma) = \frac{1}{2}\sqrt{\frac{\gamma}{\pi z}} \exp\left(-\gamma\left(\frac{\sqrt{z}-a}{\sigma}\right)^2\right) \tag{4}$$

Next, the SINR distribution in the strong NLoS environment is expressed as an exponential distribution, shown in Eq. 5, by transforming the Rayleigh distribution in Eq. 3.

$$p(z; \gamma) = \gamma \exp(-\gamma z) \tag{5}$$

Here, z is the SINR on a linear scale, and γ is a hyperparameter that determines the shape of the distribution.

Thus, we define the channel distribution for the weak NLoS environment as the Gaussian-Gamma distribution in Eq. 4 and the channel distribution for the strong NLoS environment as the exponential distribution in Eq. 5.

B. WEAK NLOS ENVIRONMENT

The architecture of the proposed method under a weak NLoS environment is illustrated in Fig. 2. This method introduces two novel modules: a Weak NLoS Detector and a Max cwnd Prediction Mechanism. The entire scheme is implemented solely by modifying the receiver-side architecture of the user equipment (UE).

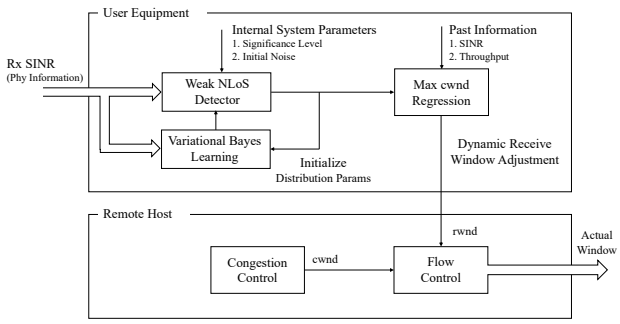

FIGURE 2. Proposed Method for Weak NLoS Environment. The proposed method involves implementing new components solely on the mobile terminal.

First, the UE measures the received SINR (Rx SINR) internally. This information is input to the Weak NLoS Detector, which detects abrupt environmental transitions—such as those from LoS to NLoS (or vice versa)—and outputs the predicted average SINR under the current wireless conditions. Variations in the wireless environment at the UE side are learned using variational Bayesian inference, enabling accurate SINR prediction even under dynamic conditions. The predicted SINR is then fed into the Max cwnd Prediction Mechanism, which estimates the maximum value of the cwnd via online regression. The predicted maximum cwnd

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

is transmitted to the sender, i.e., the Remote Host (RH), as the rwnd, enabling dynamic adjustment of the transmission window through flow control mechanisms.

Detailed descriptions of the two newly introduced modules are provided in the following paragraphs.

The Weak NLoS Detector is a mechanism that rapidly learns SINR variations. Specifically, the channel is modeled using a Gaussian-Gamma distribution, as expressed in Eq. 4, and a hypothesis test is conducted to determine whether the received SINR follows the model distribution. The null hypothesis assumes that the current weak NLoS environment follows the model distribution, while the alternative hypothesis assumes otherwise. When the significance level is set to ϵ, the test determines whether z falls within the range defined by Eq. 6.

$$z \in \left[\left(\frac{1}{\sqrt{2\rho}\sigma} \phi \left(\frac{1}{2} \pm \frac{\varepsilon - 1}{2} \right) + \frac{\alpha}{\sigma} \right)^2 \right] \tag{6}$$

Here, $\phi(x)$ represents the lower x-percentile of the standard normal distribution. The distribution parameters are updated through variational Bayesian inference based on Eqs. 7-12, which are interdependent and solved iteratively using an approximation method.

$$\alpha \leftarrow \frac{\left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} \sigma \sum_{i=1}^{m} \sqrt{z_i} + \rho\alpha}{m \left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} + \rho} \tag{7}$$

$$\rho \leftarrow m \left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} + \rho \tag{8}$$

$$k \leftarrow k + \frac{m}{2} \tag{9}$$

$$\lambda \leftarrow \lambda + \sum_{i=1}^{m} \left\langle \left(\frac{a}{\sigma} - \sqrt{z_i} \right)^2 \right\rangle_{r(a)} \tag{10}$$

$$\left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} \leftarrow \frac{1}{\sigma^2} \cdot \frac{k + \frac{m}{2}}{\lambda + \sum_{i=1}^{m} \left\langle \left(\frac{a}{\sigma} - \sqrt{z_i} \right)^2 \right\rangle_{r(a)}} \tag{11}$$

$$\sum_{i=1}^{m} \left\langle \left(\frac{a}{\sigma} - \sqrt{z_i} \right)^2 \right\rangle_{r(a)} \leftarrow \frac{1}{\sigma^2} \cdot \frac{m}{2 \left(m \left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} + \rho \right)} + \frac{1}{\sigma^2} \sum_{i=1}^{m} \left(\frac{\left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} \sigma \sum_{i=1}^{m} \sqrt{z_i} + \rho\alpha}{m \left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} + \rho} - \sigma\sqrt{z_i} \right)^2 \tag{12}$$

If the received SINR does not fall within the range specified by Eq. 6, the Weak NLoS Detector concludes that the channel environment has changed abruptly and resets the learned distribution parameters. It then starts re-learning the parameters to adapt to the new environment, where α and γ are initialized according to Eq. 13 and Eq. 14, respectively. The value of α, computed using Eq. 13, is then converted to a logarithmic scale and passed as the predicted SINR to the Max cwnd Prediction Mechanism.

$$a = \frac{\sigma}{m} \sum_{i=1}^{m} \sqrt{z_i} \tag{13}$$

$$\gamma = \frac{m}{2 \sum_{i=1}^{m} \left(\sqrt{z_i} - \frac{a}{\sigma} \right)^2} \tag{14}$$

The Max cwnd Prediction Mechanism aims to prevent excessive buffering at the base station by predicting an upper bound for cwnd. The regression model for estimating the maximum cwnd is defined by Eq. 15, assuming a proportional relationship between SINR and throughput.

$$\text{Max cwnd} = \lfloor \frac{\beta}{\text{SegmentSize}} \text{SINR} \rfloor \tag{15}$$

Here, β is the learned regression coefficient, SegmentSize is the data segment size, and $\lfloor \cdot \rfloor$ is the floor function. The optimal value of β is obtained by minimizing the loss function L defined in Eq. 16.

$$L(\mathbf{x}, \mathbf{TP}(\mathbf{x}); \beta) = \frac{1}{N} \sum_{i=0}^{N} \text{ReLU}(|TP(x_i) - \beta x_i| \cdot (TP(x_i) - \beta x_i)) + \frac{C}{2} \alpha M^2 \tag{16}$$

Here, $\mathbf{x}$ represents the input vector composed of past SINR values, and $\mathbf{TP}(\mathbf{x})$ denotes the corresponding observed throughput vector. The loss function computes the error term element-wise for each pair $(x_i, TP(x_i))$, where i indexes the elements of the vectors. The function includes a rectified linear unit (ReLU), defined in Eq. 17, and incorporates a regularization term weighted by the hyperparameters C and M, which control the sensitivity to outliers.

$$\text{ReLU}(x) = \max(0, x) \tag{17}$$

Since the loss function L is a convex function with respect to β, it has a unique minimum value. Although the function is not twice differentiable, it is Lipschitz continuous and almost everywhere differentiable, ensuring stable convergence using the gradient descent method. The update of β through gradient descent is expressed in Eqs. 18 and 19.

$$\beta \leftarrow \beta - \rho \frac{\partial L}{\partial \beta} \tag{18}$$

$$\frac{\partial L}{\partial \beta} = -2 \frac{1}{N} \sum_{i=0}^{N} x_i \left(TP(x_i) - \beta x_i \right) H \left(TP(x_i) - \beta x_i \right) + \frac{C}{2} M^2 \tag{19}$$

where the Heaviside function is defined as:

$$H(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases} \tag{20}$$

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

An online learning algorithm is also integrated into the Max cwnd Prediction Mechanism. The regression coefficient β is updated using Eq. 21, where β_{online} is the coefficient learned from the most recent N data points.

$$\beta \leftarrow w\beta + (1-w)\beta_{online} \tag{21}$$

Here, w is the weighting factor for the moving average. The value of β obtained from the minimization of Eq. 16 is substituted into Eq. 15 to calculate the maximum cwnd using the predicted SINR from the Weak NLoS Detector.

The detailed algorithm of the proposed method for Weak NLoS is illustrated in the form of pseudocode in Algorithm 1.

C. STRONG NLOS ENVIRONMENT

The architecture of the proposed method in a strong NLoS environment is shown in Fig. 3. This method introduces three new components: a Strong NLoS Detector and a Strong NLoS Verifier, both implemented on the UE, and a Congestion Control Algorithm (CCA) implemented on the RH to suppress erroneous RTO events.

As in the weak NLoS case, the UE internally measures the Rx SINR. This information is input to the Strong NLoS Detector, which identifies severe SINR degradation and recovery from such conditions. The wireless environment dynamics at the UE are learned through Bayesian learning, enabling adaptive detection under varying channel conditions. The detection result is sent to the CCA, which prevents the cwnd from being reset by RTO while the channel is classified as being in a strong NLoS state.

However, it is possible for the Strong NLoS Detector to mistakenly classify the channel as strong NLoS even when the SINR level is sufficient for data transmission. To mitigate such misclassifications, the Strong NLoS Verifier monitors the reception of acknowledge (ACK) packets while the channel is labeled as strong NLoS and adaptively adjusts the detector's parameters to refine the classification criteria.

Detailed descriptions of the three newly introduced modules are provided in the following paragraphs.

The Strong NLoS Detector learns the SINR threshold at which data transmission is interrupted, preventing misclassification of SINR degradation as congestion. First, the detector acquires the Rx SINR traced by the UE, converts it from logarithmic to linear scale, and tests whether the obtained value z follows the strong NLoS distribution defined in Eq. 5. The null hypothesis assumes that the channel follows the strong NLoS distribution, while the alternative hypothesis assumes otherwise. When the significance level is set to ϵ, the hypothesis test is performed according to Eq. 22.

$$z \geq \lambda\left(\varepsilon^{-\frac{1}{k}} - 1\right) \tag{22}$$

If z satisfies Eq. 22, the null hypothesis is rejected, indicating that the channel is in a LoS state. Otherwise, the channel is determined to be in a NLoS state. In this equation, k and λ are distribution parameters updated sequentially through Bayesian learning based on previously acquired SINR values z_i, as defined in Eqs. 23 and 24.

$$k \leftarrow k + m \tag{23}$$

$$\lambda \leftarrow \lambda + \sum_{i=1}^{m} z_i \tag{24}$$

Algorithm 1 Proposed Method for Weak NLoS.

Require: **SINR** $\in \mathbb{R}^m$, $\epsilon \in (0,1)$, $\sigma > 0$, SegmentSize $\in \mathbb{N}$, **Throughput** $\in \mathbb{R}^{+N}$, Upper_limit $\in \mathbb{N}$, $w \in (0,1)$
Ensure: rwnd $\in \mathbb{R}$
{ Weak NLoS Detector }
sinr $\leftarrow$ The most recent information in **SINR**
$z \leftarrow 10^{\frac{\text{sinr}-30}{10}}$
if z satisfies eq. 6 **then**
  distribution_change $\leftarrow$ True
else
  distribution_change $\leftarrow$ False
end if
if first step or distribution_change == True **then**
  $a \leftarrow$ eq. 13
  $\gamma \leftarrow$ eq.14
  $\alpha \leftarrow a$
  ρ, k, λ initialize
  $\left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} \leftarrow \gamma$
  $\sum_{i=1}^{m} \left\langle \left(\frac{a}{\sigma} - \sqrt{z_i}\right)^2 \right\rangle_{r(a)} \leftarrow \frac{m}{2\gamma}$
  Pre_SINR $= 10\log_{10}\left(\frac{a}{\sigma}\right)^2 + 30$
else
  while Until the parameters converge through variational Bayesian learning **do**
    $\alpha \leftarrow$ eq. 7
    $\rho \leftarrow$ eq. 8
    $k \leftarrow$ eq. 9
    $\lambda \leftarrow$ eq. 10
    $\left\langle \frac{\gamma}{\sigma^2} \right\rangle_{r(\gamma)} \leftarrow$ eq. 11
    $\sum_{i=1}^{m} \left\langle \left(\frac{a}{\sigma} - \sqrt{z_i}\right)^2 \right\rangle_{r(a)} \leftarrow$ eq. 12
  end while
end if
{ Max cwnd Prediction Mechanism }
if N unlearned data samples are collected **then**
  $\beta_{online} \leftarrow \text{argmin}_{\beta_{online}} L(\mathbf{SINR}, \mathbf{Throughput})$
  $\beta \leftarrow$ Eq. 21
end if
Max_cwnd $\leftarrow \frac{\beta}{\text{SegmentSize}}$Pre_SINR
if Max_cwnd > Upper_limit **then**
  Max_cwnd $\leftarrow$ Upper_limit
end if
{ Dynamic Receive Window Adjustment }
rwnd $\leftarrow$ Max_cwnd

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

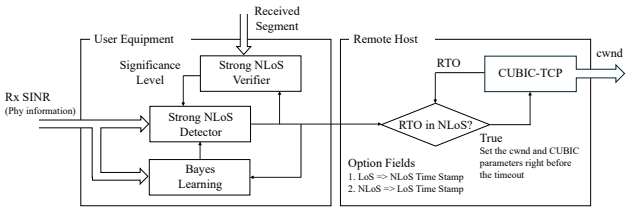

FIGURE 3. Proposed Method for Strong NLoS Environment. The proposed method requires the implementation of new components on both the sender and receiver sides, but does not rely on any proxy or intermediary devices.

Based on the test result, the Strong NLoS Detector outputs the transition time between LoS and NLoS states. This information is stored in an option field and transmitted to the RH side.

The Strong NLoS Verifier enhances the accuracy of the detector by minimizing misclassification. The verifier dynamically adjusts the significance level to reduce misclassification caused by temporary SINR degradation. If an ACK is received while the channel is classified as NLoS, the significance level ϵ is adjusted according to Eq. 25.

$$\varepsilon \leftarrow \varepsilon + \rho(1 - \sigma(z))\varepsilon^{-1} \tag{25}$$

Here, ρ is the learning rate for gradient descent, and $\sigma(z)$ is the sigmoid function defined in Eq. 26.

$$\sigma(z) = \frac{1}{1 + \exp(-\beta z)} \tag{26}$$

The significance level increases each time a misclassification occurs, making the NLoS classification more conservative and improving accuracy over time.

The CCA is designed to prevent incorrect RTO events and is based on the TCP-CUBIC algorithm. When an RTO occurs on the RH side, the occurrence time, the cwnd just before the RTO, and the CUBIC parameter K are recorded. When an ACK is received from the UE, the cause of the RTO is evaluated based on the occurrence time. If the RTO occurred within the option field period or within t seconds thereafter, the RH restores the cwnd and CUBIC parameter K to the state before the RTO event. Here, the parameter t is especially effective when ACK packets are received with delay. Under strong NLoS conditions, severe signal attenuation may prevent ACK packets from successfully reaching the RH. In such cases, the ACK packets are received after a delay once the link recovers to a LoS state. Therefore, by incorporating the parameter t, the proposed method can accurately evaluate RTO events caused by these delayed ACKs.

The implementation of the proposed method for strong NLoS is illustrated separately for the UE side and the RH side in the form of pseudocode in Algorithm 2 and Algorithm 3, respectively. The radio environment evaluation on the UE side outputs two fields: LoS_to_NLoS, indicating the timestamp when the state switches from LoS to NLoS, and NLoS_to_LoS, indicating the timestamp for the reverse transition. These fields are used on the RH side to assess whether an RTO event was caused by a strong NLoS condition. Based on this evaluation, the final cwnd value is determined and output.

Algorithm 2 Proposed Method for Strong NLoS on the UE side.

Require: SINR $\in \mathbb{R}^m$, $\epsilon \in (0, 1)$
Ensure: LoS_to_NLoS, NLoS_to_LoS
{Strong NLoS Detector}
$z \leftarrow 10^{\frac{\text{SINR}-30}{10}}$
if z satisfies eq. 22 **then**
  state ← LoS
else
  state ← NLoS
end if
if The state transitions from LoS to NLoS **then**
  LoS_to_NLoS ← the corresponding timestamp
end if
if The state transitions from NLoS to LoS **then**
  NLoS_to_LoS ← the corresponding timestamp
end if
{ Bayesian learning }
$k \leftarrow$ eq. 23
$\lambda \leftarrow$ eq. 24
{Strong NLoS Verifier}
if Receive Ack Packet and state == NLoS **then**
  $\epsilon \leftarrow$ eq. 25
  if $\epsilon > 1$ **then**
    $\epsilon \leftarrow 1$
  end if
end if

V. EVALUATION

The simulation environment is implemented using ns-3 [17]. Additionally, the mmWave module for ns-3 [18] is introduced to virtually reproduce the mmWave communication environment and evaluate the proposed congestion control method. The evaluation metrics are throughput and RTT. In multi-connection scenarios, Jain's Fairness Index is calculated when all transmitters (RHs) employ the same congestion control algorithm—either the proposed method or CUBIC TCP. All simulations described below on ns-3 are conducted 10 times independently.

A. WEAK NLOS ENVIRONMENT

First, we evaluate the regression line used in the Max cwnd Prediction Mechanism. The online regression coefficient β_{online} is trained using the most recent 10 data points, and the coefficient β is sequentially updated according to Eq. 21. The learning rate ρ is varied among 0.01, 0.1, 1, and 5, and the convergence behavior of the learned regression coefficient β is evaluated for each case.

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

Algorithm 3 Proposed Method for Strong NLoS on the RH side.

Require: LoS_to_NLoS, NLoS_to_LoS, $t \in \mathbb{R}^+$
Ensure: cwnd

```
if RTO occuers then
    timestamp ← the corresponding timestamp
    cwnd_tmp ← the cwnd of CUBIC-TCP just before the RTO
    K_tmp ← the CUBIC parameter K just before the RTO
end if
if Receive an ACK and timestamp ∈ { LoS_to_NLoS, NLoS_to_LoS + t } then
    Set cwnd_tmp in CUBIC-TCP
    Set K_tmp in CUBIC-TCP
end if
cwnd ← the cwnd of CUBIC-TCP
```

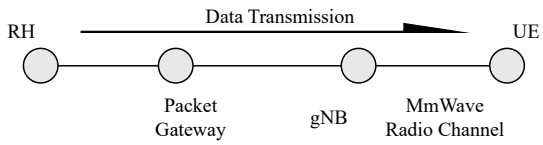

FIGURE 4. Evaluation topology for a single connection environment. This topology is used to directly observe the impact of channel variations in mmWave communication.

Next, we conduct evaluations in both single-connection and multi-connection environments.

In the single-connection scenario, we use the topology shown in Fig. 4. A single-obstacle channel model shown in Fig. 6 is used for the mmWave wireless channel to directly evaluate the bufferbloat reduction effect of the proposed method. The transmission round-trip propagation time (RTprop) is fixed at 0.12 seconds, and the initial UE positions P are set to $(100, -2, 1.8)$, $(150, -2, 1.8)$, and $(200, -2, 1.8)$ to simulate different SINR levels. After the simulation starts, the UEs move at a speed of $2\,\mathrm{m/s}$, which is approximately equivalent to a pedestrian walking speed. The congestion control algorithm is set to CUBIC and the proposed method, and the latency characteristics are compared for each case.

In the multi-connection scenario, we adopt the topology shown in Fig. 5. A large-scale NLoS channel model shown in

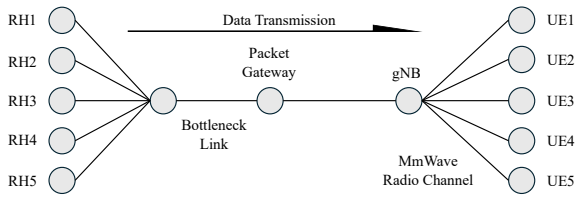

FIGURE 5. Evaluation topology for a multi-connection environment. This topology is employed to observe the combined effects of channel variations in mmWave communication and network congestion.

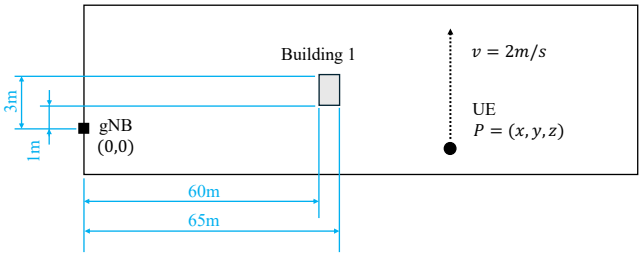

FIGURE 6. Single-obstacle Channel Model. This channel model is employed as the most basic configuration for directly analyzing the impact of NLoS conditions in a controlled environment.

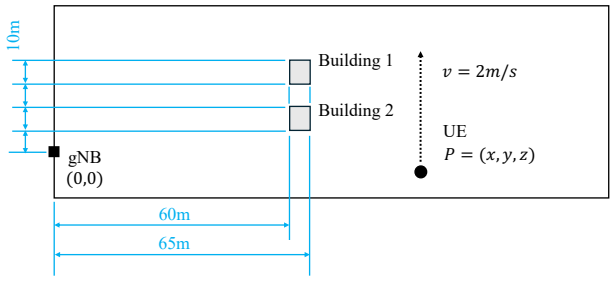

FIGURE 7. Large-scale NLoS Channel Model. This channel model is employed as a simplified setting to observe the long-term effects of multiple NLoS conditions and multipath propagation.

Fig. 7 is used for the mmWave wireless channel to evaluate the overall impact of the proposed method on communication quality. The bottleneck link bandwidth is set to 100 Gbps and 400 Mbps to simulate different congestion environments. The RTprop is fixed at 0.12 seconds, and the initial position of each UE i $(i = 1, 2, ..., 5)$ is set to $P_i = (90 + 10i, -2, 1.8)$. The moving speed is set to $2\,\mathrm{m/s}$. Throughput and Jain's Fairness Index are used as evaluation metrics. The effectiveness of the proposed method is verified by comparing the evaluation results with those obtained using CUBIC.

The remaining key parameters used in the simulations are summarized in Table 2.

TABLE 2. Simulation Parameters in Weak NLoS.

Parameter	Explanation	Value
RlcAmEnabled	Enable/Disable RLC AM Retransmission.	true
HarqEnabled	Enable/Disable HARQ.	true
SegmentSize	Segment size of transmitted packets.	2500
σ	The variance-related parameter in Eq. 4.	0.001
ϵ	The significance level used for the test in Eq. 6.	0.95
m	Number of samples required for online learning of the SINR distribution.	10
C	The regularization parameter in Eq. 16.	0.06
M	The regularization parameter in Eq. 16.	50
w	The weighting factor for the moving average in Eq. 21.	0.9
Upper_limit	The maximum rwnd value determined by the Max cwnd Prediction Mechanism.	8000

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

TABLE 3. Simulation Parameters in Strong NLoS.

Parameter	Explanation	Value
RlcAmEnabled	Enable/Disable RLC AM Retransmission.	true
HarqEnabled	Enable/Disable HARQ.	true
SegmentSize	Segment size of transmitted packets.	2500
ϵ	The significance level used for the test in Eq. 22.	0.95
m	Number of samples required for online learning of the SINR distribution.	10
β	The inverse temperature parameter in Eq. 26.	10

TABLE 4. The Average Learning Time of β_{online} for Different Learning Rates ρ.

Value of ρ	The Average Learning Time (µs)
0.01	6.76×10^3
0.1	6.72×10^2
1	6.32×10^1

B. STRONG NLOS ENVIRONMENT

First, the evaluation is conducted in a single-connection environment using the topology shown in Fig. 4. A single-obstacle channel model shown in Fig. 6 is used for the mmWave wireless channel to directly evaluate the effect of the proposed method. The initial UE position P is set to $(200, -2, 1.8)$, and the moving speed is set to $2\,\mathrm{m/s}$. The RTprop is varied at 0.04 seconds, 0.12 seconds, and 0.20 seconds to simulate different latency conditions. The congestion control algorithm applies the proposed method with the CCA module parameter t set to $t = 0$ and $t = 0.5$. The evaluation metrics are throughput and RTT. The effectiveness of the proposed method is evaluated by comparing the results with those obtained using CUBIC.

Next, the evaluation is conducted in a multi-connection environment using the topology shown in Fig. 5. A large-scale NLoS channel model shown in Fig. 7 is used for the mmWave wireless channel to evaluate the effect of the Strong NLoS Verifier in the proposed method. The bottleneck link bandwidth is set to 400 Mbps, and the RTprop is varied at 0.04 seconds, 0.12 seconds, and 0.20 seconds to simulate different latency conditions under congestion. The initial position of each UE i $(i = 1, 2, ..., 5)$ is set to $P_i = (90 + 10i, -2, 1.8)$, and the moving speed is set to $2\,\mathrm{m/s}$. The congestion control algorithm applies the proposed method with the Strong NLoS Verifier disabled ($\rho = 0$) and enabled ($\rho = 0.0002$).

To evaluate the impact of the Strong NLoS Verifier on communication quality, the minimum throughput and minimum Jain's Fairness Index [19] are compared among the results from 10 simulation runs. The remaining key parameters used in the simulations are summarized in Table 3.

VI. RESULTS

A. WEAK NLOS ENVIRONMENT

Fig. 8 depicts the trajectory of the regression coefficient β learned with a learning rate of $\rho = 0.01$. The results indicate that β converges near 0.6, demonstrating that the online learning based on the minimization of Eq. 16 operates correctly. Nearly identical convergence behavior—so close that including separate graphs would be redundant—is observed for learning rates of $\rho = 0.1$ and $\rho = 1$. However, when the learning rate is increased to $\rho = 5$, divergence occurs, and the learning process fails to converge. Table 4 summarizes the average update time of β_{online} for each learning rate. Within the convergent range of β, it is evident that increasing the learning rate significantly reduces the learning time.

Fig. 9 shows the channel learning results in the weak NLoS

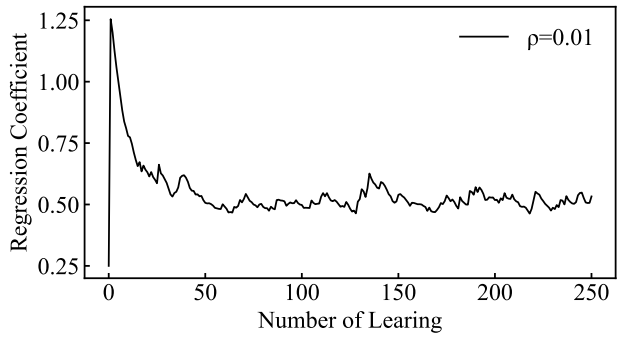

FIGURE 8. Results of Online Learning of the Regression Coefficient β.

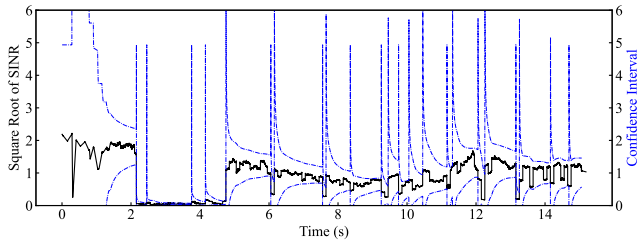

FIGURE 9. Learning Results of the Channel in Weak NLoS Environment.

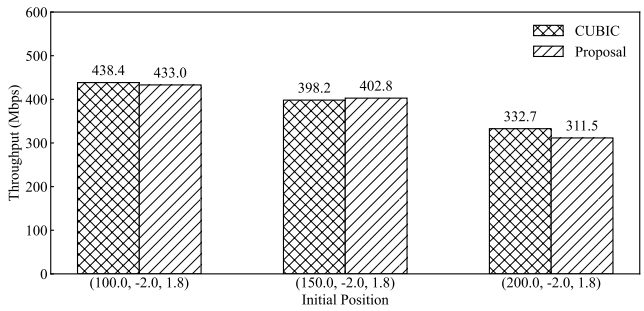

FIGURE 10. Throughput Evaluation Results of CUBIC and the Proposed Method in a Single-Connection Environment.

TABLE 5. Throughput and Fairness Comparison of CUBIC and the Proposed Method in a Non-Congestion Scenario.

Congestion Control	Throughput (Mbps)	Jain's Fairness Index
CUBIC	67.7612	0.9840
Proposal	74.3891	0.9848

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

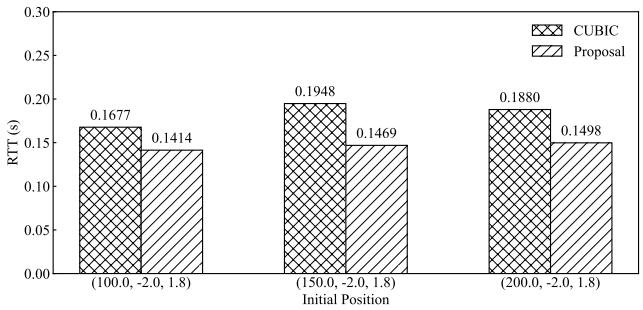

FIGURE 11. RTT Evaluation Results of CUBIC and Proposed Method in a Single-Connection Environment.

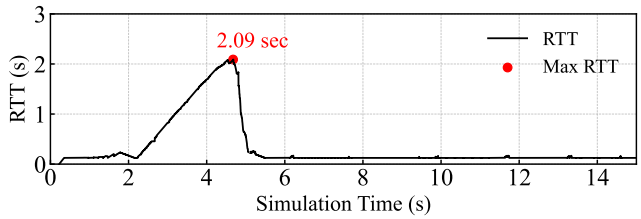

FIGURE 12. Time-Series Variation of RTT in CUBIC in a Single-Connection Environment.

environment. In this figure, the blue dashed lines represent the upper and lower bounds of the confidence interval. From the results, it can be confirmed that the weak NLoS classifier efficiently learns the SINR distribution and performs weak NLoS testing with an appropriate confidence interval width. Furthermore, the average computation time for a single parameter update in variational Bayesian learning is 6.41 µs, indicating that channel learning can be performed in near real-time.

Next, the communication quality evaluation in a single-connection environment is presented. Fig. 10 shows the

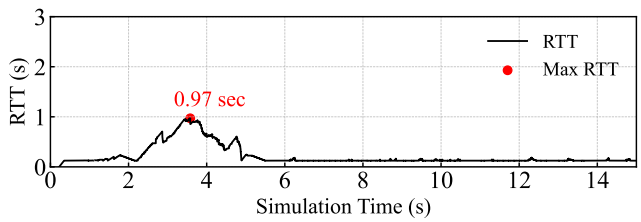

FIGURE 13. Time-Series Variation of RTT in Proposed Method in a Single-Connection Environment.

TABLE 6. Throughput and Fairness Comparison of CUBIC and the Proposed Method in a Congestion Scenario.

Congestion Control	Throughput (Mbps)	Jain's Fairness Index
CUBIC	61.7903	0.9281
Proposal	60.6217	0.9513

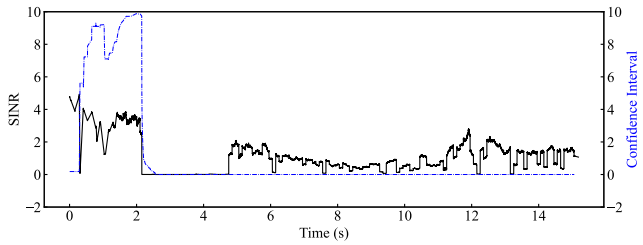

FIGURE 14. Learning Results of the Channel in Strong NLoS Environment.

TABLE 7. Evaluation Results of the Minimum Throughput in the Proposed Method.

Round-trip Propagation Time (s)	Without Strong NLoS Verifier (Mbps)	With Strong NLoS Verifier (Mbps)	Improvement Rate (%)
0.04	60.1002	58.7665	-2.2
0.12	58.0737	58.3096	+0.4
0.20	52.2955	57.9800	**+11.0**

throughput evaluation results. The results indicate that no significant difference in throughput is observed between CUBIC and the proposed method in all scenarios. Furthermore, a t-test on the mean throughput difference between CUBIC and the proposed method are conducted for each scenario illustrated in Fig. 10 at a 5% significance level, revealing no statistically significant difference.

Fig. 11 shows the evaluation results for the average RTT. The results indicate that the average RTT decreases in all scenarios. Furthermore, a t-test conducted on the average RTT, similar to the throughput, reveals a significant difference.

A detailed comparison of the time-series variation of RTT between CUBIC and the proposed method is shown. Fig. 12 presents the time-series variation of RTT when using CUBIC, and Fig. 13 presents the time-series variation of RTT when using the proposed method. The results show that, particularly, the maximum RTT is 2.09 seconds for CUBIC and 0.97 seconds for the proposed method, achieving a 53.6% reduction. These results confirm that the proposed method can mitigate bufferbloat.

Next, the communication quality evaluation in a multi-connection environment is presented. For the non-congestion scenario, Table 5 shows the throughput and fairness evaluation results. The results indicate that, compared to CUBIC, the proposed method improves the average throughput of the entire UE by 9.8%. Additionally, the Jain's Fairness Index is 0.9840 for CUBIC and 0.9848 for the proposed method, showing that both methods are almost identical in fairness.

For the congestion scenario, Table 6 shows the throughput and fairness evaluation results. The results indicate that there is no significant difference in the average throughput of the entire UE, while fairness improves by 0.023. These results suggest that the proposed method can maintain communication quality equal to or better than CUBIC.

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

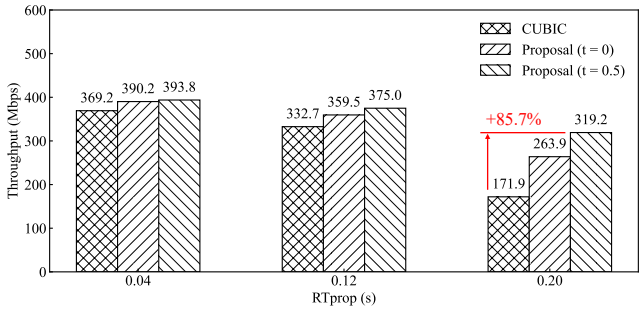

FIGURE 15. Throughput Evaluation Results of CUBIC and Proposed Method.

TABLE 8. Minimum Fairness Evaluation Results of Proposed Method.

Round-trip Propagation Time (s)	Without Strong NLoS Verifier (Mbps)	With Strong NLoS Verifier (Mbps)	Improvement Rate (%)
0.04	0.8363	0.7812	-6.6
0.12	0.7344	0.9240	+25.8
0.20	0.6040	0.8514	**+41.0**

B. STRONG NLOS ENVIRONMENT

The learning results of the channel are shown in Fig.14. In this figure, the blue dashed line represents the upper bound of the confidence interval. From Fig.14, it can be observed that the strong NLoS environment is learned rapidly, approximately two seconds after the simulation begins. Additionally, the average computation time required for a single parameter update in Bayesian learning is $0.191\,\mu s$, demonstrating that channel learning can be performed in near real-time.

The communication performance evaluation in the single-connection scenario is presented below. Fig. 15 shows the throughput results. From the results, it can be seen that throughput improves in the order of CUBIC, the proposed method ($t = 0$), and the proposed method ($t = 0.5$) under all evaluated RTT conditions. In particular, when RTprop is 0.20 seconds, the proposed method with $t = 0$ improves throughput by 53.6% compared to CUBIC, while the proposed method with $t = 0.5$ achieves an 85.7% improvement.

Next, the effectiveness of the Strong NLoS Verifier in the multi-connection scenario is evaluated. Table 7 shows the minimum throughput obtained from 10 simulation runs under each RTT condition. The results indicate that, when the Strong NLoS Verifier is disabled, throughput tends to decrease as RTprop increases. In contrast, when the verifier is enabled, throughput remains stable. Notably, when RTprop is 0.20 seconds, enabling the NLoS verifier results in an 11.0% improvement in throughput.

Table 8 presents the minimum Jain's Fairness Index recorded under each RTT condition. The results show that enabling the Strong NLoS Verifier significantly improves fairness for RTprop values of 0.12 seconds and 0.20 seconds. Specifically, for RTprop = 0.20 seconds, fairness improves by approximately 41.0% compared to when the verifier is disabled.

These results demonstrate that incorporating the Strong NLoS Verifier into the proposed method enhances communication quality under congested conditions.

VII. DEPLOYABILITY DISCUSSION

For the proposed algorithms to be widely adopted in real-world mmWave mobile communication systems, it is essential not only to enhance communication performance but also to ensure ease of deployment. In this section, we discuss the implementability and practical deployability of the proposed approaches from the perspective of real-world adoption.

The weak NLoS approach relies solely on local SINR measurements at the UE to infer link blockages and on dynamic window adaptation based on flow control, without requiring any additional signaling from the TCP sender or the network side, thereby preserving full backward compatibility with existing TCP implementations and network infrastructure. For instance, the cwnd can be adjusted within the UE software based on the observed SINR values. As all processing is performed locally at the UE, this approach offers a cost-effective and easily deployable solution that also enables flexible deployment, as the algorithm can be selectively enabled or disabled by the UE without any coordination with the network.

In contrast, the strong NLoS approach requires an explicit feedback channel between the UE and the server to report NLoS events, necessitating modifications to the TCP stack at both endpoints. While this variant can react more promptly to link blockages and potentially achieve higher throughput, it results in increased implementation complexity on both the end devices and the network infrastructure.

In summary, **both the weak and strong NLoS approaches are end-to-end solutions, which provide advantages over prior work in terms of deployability**. The selection between these approaches can be made based on specific application requirements, such as the trade-off between performance gain and implementation cost.

VIII. CONCLUSION

MmWave mobile communication is a key enabler for overcoming spectrum scarcity and supporting next-generation infrastructure. However, its high-frequency characteristics are fundamentally misaligned with the conventional protocol stack, especially TCP congestion control, leading to substantial performance degradation. Previous studies have neither sufficiently analyzed the root causes of this issue nor provided practical, low-cost architectural solutions for real-world deployment.

To address these challenges, this paper proposes a cross-layer TCP congestion control architecture tailored for mmWave mobile environments. By classifying NLoS conditions into two types—weak NLoS and strong NLoS—the specific communication challenges associated with each condition are systematically analyzed. Furthermore, a method is introduced to physically define these NLoS states using SINR, enabling real-time observation of channel conditions.

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

In the proposed scheme, under weak NLoS, the receiver dynamically adjusts the cwnd based on the observed SINR to prevent excessive data transmission. In contrast, under strong NLoS, the method suppresses inappropriate RTO when SINR drops significantly, thereby avoiding the undesired reduction of the cwnd.

The scheme is lightweight, end-host-only, and fully compatible with existing infrastructure (see Section VII). Extensive ns-3 simulations show that it significantly improves performance—reducing maximum RTT by 53.6% and increasing throughput by 9.8% in weak NLoS, and improving throughput by up to 85.7% in strong NLoS scenarios while maintaining fairness.

This work provides a practical, deployable solution to a fundamental transport-layer problem in mmWave communications, advancing high-speed, low-latency networks for future applications such as autonomous driving and smart cities.

IX. FUTURE WORK

Several potential directions remain unaddressed in this paper due to scope limitations. These are briefly outlined in the following subsections to guide future investigations.

A. INCREASED NETWORK TOPOLOGY COMPLEXITY

This study evaluated the proposed scheme using a simple network topology. Its impact on communication quality in more complex and realistic network environments remains unclear. Future work should assess the scheme under more intricate scenarios, such as Parking Lot and Cross Traffic topologies, to examine its robustness and practical applicability.

B. CHANNEL MODEL GENERALIZATION

This study is based on a simplified wireless channel model that does not fully reflect the complexity of real-world propagation conditions. Future evaluations should incorporate more detailed modeling, including building geometry, dynamic obstacles, and diverse UE mobility patterns, to verify the effectiveness of the proposed scheme under realistic scenarios.

C. COMPARISON WITH ALTERNATIVE CONGESTION CONTROL MECHANISMS

The evaluation in this study focuses mainly on a comparison with CUBIC, the currently dominant congestion control algorithm. For a more comprehensive assessment, future work should incorporate comparisons with modern schemes such as BBR and Copa, as well as with representative approaches used in recent related research.

D. EVALUATION USING OPEN DATASET

To ensure the reproducibility of the communication performance improvements achieved by the proposed method and to enable fair comparisons with alternative approaches, future work should include evaluations using publicly available open datasets.

E. IMPACT OF THE REGRESSION COEFFICIENT

The regression coefficient β converges to approximately 0.6 under the chosen parameter settings. However, depending on the values of hyperparameters such as C and M, β may converge to suboptimal values. Future work should investigate how such variations affect communication performance and evaluate the robustness of the proposed scheme under different convergence behaviors.

F. APPLICABILITY BEYOND MMWAVE BANDS

While the proposed scheme is designed and evaluated primarily for mmWave communications, we believe it has the potential to be adapted to other dynamic wireless environments. In particular, environments operating in high-frequency bands under weak NLoS conditions or with rich multipath propagation that degrades SINR may also benefit from the proposed approach. However, its effectiveness in such scenarios has not yet been verified, and further evaluation is required to confirm its ability to mitigate bufferbloat more effectively than conventional congestion control methods. Moreover, as bufferbloat cannot be entirely eliminated, some degree of jitter may still persist. Therefore, for ultra delay-sensitive or jitter-intolerant applications, further investigation is needed to assess the practical applicability and performance limits of the proposed scheme.

ACKNOWLEDGMENT

This work was partly supported by NICT, Grant Number **09101**, Japan, and JST ASPIRE Grant Number **JPMJAP2323** Japan.

REFERENCES

[1] R. Yongmao, Y. Wanghong, Z. Xu, C. Huan, and L. Bing, "A survey on tcp over mmwave," *Comput. Commun.*, vol. 171, pp. 80–88, Nov. 2021.

[2] R. Poorzare and A. C. Augé, "Fb-tcp: A 5g mmwave friendly tcp for urban deployments," *IEEE Access*, vol. 9, pp. 82 812–82 832, 2021.

[3] R. Poorzare and A. Calveras, "Deep learning tcp for mitigating nlos impairments in 5g mmwave," *International Journal of Computers, Communications and Control*, vol. 18, no. 4, p. 4874, Aug 2023.

[4] T. Azzino, M. Drago, M. Polese, A. Zanella, and M. Zorzi, "X-tcp: a cross layer approach for tcp uplink flows in mmwave networks," in *2017 16th Annual Mediterranean Ad Hoc Networking Workshop (Med-Hoc-Net)*, 2017, pp. 1–6.

[5] W. Na, B. Bae, S. Cho, and N. Kim, "Dl-tcp: Deep learning-based transmission control protocol for disaster 5g mmwave networks," *IEEE Access*, vol. 7, pp. 145 134–145 144, 2019.

[6] M. Polese, M. Mezzavilla, M. Zhang, J. Zhu, S. Rangan, S. Panwar, and M. Zorzi, "milliproxy: A tcp proxy architecture for 5g mmwave cellular systems," in *2017 51st Asilomar Conference on Signals, Systems, and Computers*, 2017, pp. 951–957.

[7] P. Netalkar, J. Chen, and D. Raychaudhuri, "mmcptp: A cross-layer pull based transport protocol for 5g mmwave networks," in *NOMS 2023-2023 IEEE/IFIP Network Operations and Management Symposium*, 2023, pp. 1–9.

[8] R. Poorzare and O. P. Waldhorst, "Toward the implementation of mptcp over mmwave 5g and beyond: Analysis, challenges, and solutions," *IEEE Access*, vol. 11, pp. 19 534–19 566, 2023.

[9] R. Adams, "Active queue management: A survey," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 3, pp. 1425–1476, Third Quarter 2013.

[10] K. Nichols and V. Jacobson, "Controlling queue delay: A modern aqm is just one piece of the solution to bufferbloat," *Queue*, vol. 10, no. 5, pp. 20–34, May 2012. [Online]. Available: https://doi.org/10.1145/2208917.2209336

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2025.3601170

[11] J. Bai, T. Zhang, and G. Xie, "Macc: Cross-layer multi-agent congestion control with deep reinforcement learning," 2022. [Online]. Available: https://arxiv.org/abs/2206.01972
[12] O. I. Alramli, Z. M. Hanapi, M. Othman, I. Ahmad, and N. Samian, "Rttv-tcp: Adaptive congestion control algorithm based on rtt variations for mmwave networks," *Ad Hoc Networks*, vol. 164, p. 103611, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1570870524002221
[13] H. Jiang, Y. Wang, K. Lee, and I. Rhee, "Drwa: A receiver-centric solution to bufferbloat in cellular networks," *IEEE Transactions on Mobile Computing*, vol. 15, no. 11, pp. 2719–2734, 2016.
[14] L. Brakmo and L. Peterson, "Tcp vegas: end to end congestion avoidance on a global internet," *IEEE Journal on Selected Areas in Communications*, vol. 13, no. 8, pp. 1465–1480, 1995.
[15] X. Liu, F. Ren, R. Shu, T. Zhang, and T. Dai, *3 Mitigating Bufferbloat with Receiver-based TCP Flow Control Mechanism in Cellular Networks*. River Publishers, 2017, pp. 63–90.
[16] T. R. R. Marins, A. A. D. Anjos, C. R. N. D. Silva, V. M. R. Peñarrocha, L. Rubio, J. Reig, R. A. A. De Souza, and M. D. Yacoub, "Fading evaluation in standardized 5g millimeter-wave band," *IEEE Access*, vol. 9, pp. 67 268–67 280, 2021.
[17] "NS-3," https://www.nsnam.org.
[18] NS-3 Consortium, "Wns3 2015 submission 41," 2015, accessed: 2024-12-23.
[19] R. Jain, D. Chiu, and W. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," 1998. [Online]. Available: https://arxiv.org/abs/cs/9809099

MIKITO TAKAHASHI received his B.E. degree in Systems Innovation from the Faculty of Engineering at The University of Tokyo, Japan, in 2025. His research interests focus on cross-layer network optimization, specifically congestion control mechanisms for millimeter-wave wireless communications. His undergraduate thesis introduced a novel cross-layer congestion control algorithm that significantly improves throughput, fairness, and latency performance in 5G millimeter-wave networks. He has presented his work at national conferences including RISING 2024 and IEICE HCL 2024. Takahashi was awarded the Faculty of Engineering Dean's Award for his outstanding academic achievements and recognized for obtaining the highest grade among all graduates of the department for his undergraduate thesis. He is currently preparing further publications based on his innovative contributions to wireless networking.

AKIHIRO NAKAO Akihiro Nakao received B.S. (1991) in Physics, M.E. (1994) in Information Engineering from the University of Tokyo. He was at IBM Yamato Laboratory, Tokyo Research Laboratory, and IBM Texas Austin from 1994 till 2005. He received M.S. (2001) and Ph.D. (2005) in Computer Science from Princeton University. He taught as an associate professor (2005-2014) and as a professor (2014-2021) in Applied Computer Science, at Interfaculty Initiative in Information Studies, Graduate School of Interdisciplinary Information Studies, the University of Tokyo. He has served as Vice Dean of the University of Tokyo's Interfaculty Initiative in Information Studies (2019-2021). In April 2021, he has moved to School of Engineering, the University of Tokyo (2021-present). Since April 2023, he has been serving as Head of Department of System Innovations, School of Engineering. He was appointed as an adviser to the President of the University of Tokyo (2019-2020) and has been a special adviser to the President of the University of Tokyo (2020-present). He is serving as Director, Collaborative Research Institute for NGCI,Next-Generation Cyber Infrastructure, the University of Tokyo (2021-present). He has been appointed as the first guest professor at the University of Oulu and its Faculty of Information Technology and Electrical Engineering (ITEE) in September 2023 (2023-present). For social services, he has been playing several important roles in Japanese government and also at research societies. From 2020 to present, he is a chair and advisor of IEICE technical committee on network systems (NS) as well as a chair of IEICE technical committee on cross-field research association of super-intelligent networking (RISING). He has been elected to become the president of Communication Society, IEICE, in 2024 (2024-present) and Co-Chairperson of XGMF (XG Mobile Promotion Forum (2024-present)

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/